

4 ZFS File System

disk which is in the format cXtXdXpX. If you want to know the name of disk devices attached to your system, execute `$ format -e` in a terminal. This will list out all the disks along with any removable media.

The new ZFS file system, tank, can use as much of the disk space as needed, and is automatically mounted at /tank.

```
# mkfile 100m /tank/foo
# df -h /tank
```

Filesystem	size	used	avail	capacity
Mounted on				
tank		80G	100M	80G
1% /tank.				

Within a pool, you will probably want to create additional file systems. File systems provide points of administration that allow you to manage different sets of data within the same pool. The following example illustrates how to create a file system named fs in the storage pool tank.

```
# zfs create tank/fs
```

The new ZFS file system, tank/fs, can use as much of the disk space as needed, and is automatically mounted at /tank/fs.

```
# mkfile 100m /tank/fs/foo
# df -h /tank/fs
```

Filesystem	size	used
avail capacity		Mounted on
tank/fs		80G
100M	80G	1%
/tank/fs		

4.8.2 Creating a ZFS Storage Pool

The previous example illustrates the simplicity of ZFS. The remainder of this chapter demonstrates a more complete example similar to what you would encounter in your environment. The first tasks are to identify your storage requirements and create a storage pool. The pool describes the physical characteristics of the storage and must be created before any file systems are created.

Follow the following steps to create a ZFS storage pool:

Quick Tip

You do not need a hard disk or USB devices to try these things. Just make a file and you can use it with ZFS as a storage device. In order to create a file, use the following command:
`$ mkfile <size of the file> <name of the file>`

Quick Tip

Time slider in OpenSolaris makes use of ZFS Snapshots. Therefore in the beginning, there is hardly an overhead when Time Slider creates snapshots. But if you start running out of space, then you can delete some of the old snapshots to free up space.